Table 1: Detailed instance-level results of MMO_b, MMO_o, and PMO for Next Release Problem, showing Scott-Knott ESD ranks and the mean ± standard deviation of optimized and normalized performance (smaller is better).

Instance	**Rank**			**Mean ± Std**		
	MMO_b	MMO_o	PMO	MMO_b	MMO_o	PMO
nrp-e1	1.0	1.0	2.0	0.016 ± 0.015	0.016 ± 0.015	0.749 ± 0.193
nrp-e2	1.0	1.0	2.0	0.006 ± 0.006	0.006 ± 0.006	0.593 ± 0.163
nrp-e3	1.0	1.0	2.0	0.001 ± 0.002	0.001 ± 0.002	0.489 ± 0.190
nrp-e4	1.0	1.0	2.0	0.003 ± 0.009	0.005 ± 0.015	0.769 ± 0.145
nrp-g1	1.0	1.0	2.0	0.000 ± 0.000	0.000 ± 0.000	0.619 ± 0.259
nrp-g2	1.0	1.0	2.0	0.000 ± 0.000	0.000 ± 0.000	0.689 ± 0.130
nrp-g3	1.0	1.0	2.0	0.000 ± 0.000	0.000 ± 0.000	0.580 ± 0.238
nrp-g4	1.0	1.0	2.0	0.001 ± 0.002	0.001 ± 0.002	0.721 ± 0.168
nrp-m1	1.0	1.0	2.0	0.005 ± 0.007	0.005 ± 0.007	0.779 ± 0.143
nrp-m2	1.0	1.0	2.0	0.010 ± 0.009	0.010 ± 0.009	0.865 ± 0.102
nrp-m3	1.0	1.0	2.0	0.003 ± 0.003	0.003 ± 0.003	0.612 ± 0.176
nrp-m4	1.0	1.0	2.0	0.008 ± 0.006	0.008 ± 0.006	0.762 ± 0.125
nrp1	1.0	1.0	2.0	0.000 ± 0.000	0.000 ± 0.000	0.411 ± 0.210
nrp2	1.0	2.0	3.0	0.135 ± 0.087	0.185 ± 0.110	0.657 ± 0.142
nrp3	1.0	1.0	2.0	0.007 ± 0.006	0.007 ± 0.007	0.827 ± 0.086
nrp4	1.0	1.0	2.0	0.001 ± 0.001	0.001 ± 0.001	0.933 ± 0.048
nrp5	1.0	1.0	2.0	0.000 ± 0.000	0.000 ± 0.000	0.926 ± 0.049
Average	**1.00**	**1.06**	**2.06**	**0.011 ± 0.009**	**0.015 ± 0.011**	**0.705 ± 0.151**

Table 2: Detailed instance-level results of MMO_b, MMO_o, and PMO for Software Configuration Tuning, showing Scott-Knott ESD ranks and the mean ± standard deviation of optimized and normalized performance (smaller is better).

Instance	**Rank**			**Mean ± Std**		
	MMO_b	MMO_o	PMO	MMO_b	MMO_o	PMO
dnn_adiac	2.0	2.0	1.0	0.328 ± 0.354	0.680 ± 0.353	0.204 ± 0.029
dnn_adiac_reverse	1.0	3.0	2.0	0.000 ± 0.000	0.371 ± 0.348	0.141 ± 0.121
dnn_coffee	2.0	3.0	1.0	0.000 ± 0.000	0.267 ± 0.432	0.000 ± 0.000
dnn_coffee_reverse	1.0	2.0	2.0	0.000 ± 0.000	0.000 ± 0.000	0.400 ± 0.516
dnn_dsr	1.0	2.0	3.0	0.004 ± 0.011	0.102 ± 0.198	0.526 ± 0.291
dnn_dsr_reverse	1.0	1.0	1.0	0.000 ± 0.000	0.100 ± 0.316	0.100 ± 0.316
dnn_sa	1.0	3.0	2.0	0.023 ± 0.072	0.565 ± 0.388	0.325 ± 0.071
dnn_sa_reverse	1.0	2.0	3.0	0.000 ± 0.000	0.414 ± 0.356	0.570 ± 0.396
llvm	1.0	1.0	1.0	0.500 ± 0.000	0.500 ± 0.000	0.500 ± 0.000
llvm_reverse	1.0	2.0	3.0	0.213 ± 0.115	0.551 ± 0.186	0.737 ± 0.349
lrzip	1.0	2.0	3.0	0.167 ± 0.127	0.303 ± 0.173	0.535 ± 0.313
lrzip_reverse	1.0	1.0	2.0	0.000 ± 0.000	0.000 ± 0.001	0.148 ± 0.306
mariadb	2.0	3.0	1.0	0.000 ± 0.000	0.298 ± 0.480	0.000 ± 0.000
mariadb_reverse	1.0	3.0	2.0	0.000 ± 0.000	0.656 ± 0.311	0.341 ± 0.307
mongodb	1.0	2.0	3.0	0.062 ± 0.196	0.409 ± 0.371	0.556 ± 0.196
mongodb_reverse	1.0	2.0	2.0	0.100 ± 0.316	0.900 ± 0.316	0.900 ± 0.316
storm_rs	1.0	2.0	3.0	0.000 ± 0.000	0.158 ± 0.334	0.825 ± 0.139
storm_rs_reverse	2.0	2.0	1.0	0.000 ± 0.000	0.685 ± 0.403	0.011 ± 0.036
storm_wc	1.0	3.0	2.0	0.000 ± 0.000	0.197 ± 0.347	0.100 ± 0.316
storm_wc_reverse	1.0	3.0	2.0	0.000 ± 0.000	0.625 ± 0.320	0.249 ± 0.350
trimesh	1.0	1.0	1.0	0.500 ± 0.000	0.500 ± 0.000	0.500 ± 0.000
trimesh_reverse	1.0	2.0	3.0	0.053 ± 0.044	0.279 ± 0.237	0.568 ± 0.340
vp9	1.0	2.0	3.0	0.000 ± 0.000	0.155 ± 0.343	0.487 ± 0.370
vp9_reverse	1.0	2.0	3.0	0.045 ± 0.096	0.213 ± 0.172	0.472 ± 0.257
x264	1.0	1.0	1.0	0.500 ± 0.000	0.500 ± 0.000	0.500 ± 0.000
x264_reverse	1.0	1.0	1.0	0.500 ± 0.000	0.500 ± 0.000	0.500 ± 0.000
Average	**1.15**	**2.04**	**2.00**	**0.115 ± 0.051**	**0.382 ± 0.246**	**0.392 ± 0.205**

Table 3: Detailed instance-level results of $\mathrm{MMO_b}$, $\mathrm{MMO_o}$, and PMO for Software Effort Estimation, showing Scott-Knott ESD ranks and the mean ± standard deviation of optimized and normalized performance (smaller is better).

Instance	**Rank**			**Mean ± Std**		
	$\mathrm{MMO_b}$	$\mathrm{MMO_o}$	PMO	$\mathrm{MMO_b}$	$\mathrm{MMO_o}$	PMO
china_fold1	1.0	3.0	2.0	0.097 ± 0.066	0.373 ± 0.256	0.167 ± 0.071
china_fold2	1.0	3.0	2.0	0.027 ± 0.032	0.542 ± 0.314	0.046 ± 0.033
china_fold3	1.0	3.0	2.0	0.106 ± 0.116	0.386 ± 0.318	0.232 ± 0.128
desharnais_fold1	2.0	2.0	1.0	0.239 ± 0.225	0.444 ± 0.311	0.205 ± 0.162
desharnais_fold2	1.0	3.0	2.0	0.135 ± 0.103	0.461 ± 0.252	0.232 ± 0.160
desharnais_fold3	1.0	1.0	2.0	0.139 ± 0.105	0.152 ± 0.124	0.443 ± 0.270
finnish_fold1	1.0	3.0	2.0	0.253 ± 0.185	0.517 ± 0.206	0.416 ± 0.305
finnish_fold2	1.0	2.0	2.0	0.087 ± 0.102	0.203 ± 0.174	0.288 ± 0.360
finnish_fold3	1.0	3.0	2.0	0.050 ± 0.039	0.350 ± 0.253	0.178 ± 0.100
maxwell_fold1	1.0	2.0	2.0	0.189 ± 0.157	0.414 ± 0.166	0.441 ± 0.264
maxwell_fold2	1.0	3.0	2.0	0.222 ± 0.144	0.738 ± 0.225	0.400 ± 0.142
maxwell_fold3	1.0	3.0	2.0	0.202 ± 0.171	0.679 ± 0.163	0.614 ± 0.202
miyazaki_fold1	1.0	3.0	2.0	0.118 ± 0.072	0.439 ± 0.291	0.229 ± 0.134
miyazaki_fold2	1.0	3.0	2.0	0.174 ± 0.116	0.480 ± 0.217	0.387 ± 0.233
miyazaki_fold3	1.0	3.0	2.0	0.222 ± 0.119	0.502 ± 0.242	0.386 ± 0.241
Average	**1.07**	**2.67**	**1.93**	**0.151 ± 0.117**	**0.445 ± 0.234**	**0.311 ± 0.187**

Table 4: Detailed instance-level results of $\mathrm{MMO_b}$, $\mathrm{MMO_o}$, and PMO for Software Defect Prediction, showing Scott-Knott ESD ranks and the mean ± standard deviation of optimized and normalized performance (smaller is better).

Instance	**Rank**			**Mean ± Std**		
	$\mathrm{MMO_b}$	$\mathrm{MMO_o}$	PMO	$\mathrm{MMO_b}$	$\mathrm{MMO_o}$	PMO
ant-1.7_J48	1.0	2.0	3.0	0.272 ± 0.181	0.485 ± 0.235	0.582 ± 0.207
ant-1.7_KNN	1.0	2.0	2.0	0.417 ± 0.212	0.653 ± 0.206	0.593 ± 0.187
ant-1.7_LR	1.0	1.0	1.0	0.392 ± 0.299	0.429 ± 0.307	0.408 ± 0.304
ant-1.7_NB	1.0	3.0	2.0	0.414 ± 0.223	0.555 ± 0.303	0.468 ± 0.264
camel-1.6_J48	1.0	2.0	3.0	0.321 ± 0.279	0.473 ± 0.308	0.600 ± 0.373
camel-1.6_KNN	1.0	2.0	3.0	0.283 ± 0.142	0.354 ± 0.181	0.571 ± 0.214
camel-1.6_LR	1.0	2.0	2.0	0.413 ± 0.268	0.747 ± 0.237	0.674 ± 0.343
camel-1.6_NB	1.0	2.0	2.0	0.393 ± 0.288	0.559 ± 0.292	0.592 ± 0.312
ivy-2.0_J48	1.0	2.0	3.0	0.533 ± 0.261	0.620 ± 0.304	0.650 ± 0.304
ivy-2.0_KNN	1.0	2.0	2.0	0.498 ± 0.260	0.575 ± 0.250	0.611 ± 0.221
ivy-2.0_LR	1.0	3.0	2.0	0.601 ± 0.267	0.723 ± 0.258	0.680 ± 0.243
ivy-2.0_NB	1.0	2.0	2.0	0.646 ± 0.280	0.687 ± 0.286	0.673 ± 0.286
jedit-4.0_J48	1.0	1.0	2.0	0.492 ± 0.303	0.543 ± 0.342	0.568 ± 0.359
jedit-4.0_KNN	1.0	2.0	2.0	0.420 ± 0.182	0.523 ± 0.185	0.586 ± 0.210
jedit-4.0_LR	1.0	3.0	2.0	0.433 ± 0.307	0.618 ± 0.302	0.575 ± 0.278
jedit-4.0_NB	1.0	2.0	2.0	0.525 ± 0.275	0.588 ± 0.282	0.596 ± 0.302
lucene-2.4_J48	1.0	2.0	3.0	0.368 ± 0.203	0.546 ± 0.245	0.713 ± 0.224
lucene-2.4_KNN	1.0	2.0	3.0	0.444 ± 0.241	0.551 ± 0.207	0.651 ± 0.320
lucene-2.4_LR	1.0	2.0	2.0	0.388 ± 0.276	0.582 ± 0.256	0.537 ± 0.251
lucene-2.4_NB	1.0	2.0	2.0	0.401 ± 0.236	0.492 ± 0.254	0.465 ± 0.255
poi-3.0_J48	1.0	2.0	3.0	0.373 ± 0.245	0.456 ± 0.245	0.546 ± 0.314
poi-3.0_KNN	1.0	2.0	2.0	0.601 ± 0.271	0.733 ± 0.290	0.729 ± 0.285
poi-3.0_LR	1.0	3.0	2.0	0.634 ± 0.270	0.781 ± 0.208	0.682 ± 0.253
poi-3.0_NB	1.0	2.0	2.0	0.569 ± 0.261	0.674 ± 0.214	0.623 ± 0.241
synapse-1.2_J48	1.0	2.0	2.0	0.344 ± 0.280	0.528 ± 0.286	0.586 ± 0.291
synapse-1.2_KNN	1.0	2.0	3.0	0.382 ± 0.276	0.475 ± 0.217	0.516 ± 0.234
synapse-1.2_LR	1.0	2.0	2.0	0.462 ± 0.254	0.557 ± 0.253	0.547 ± 0.302
synapse-1.2_NB	1.0	2.0	2.0	0.493 ± 0.295	0.546 ± 0.284	0.542 ± 0.274
velocity-1.6_J48	1.0	2.0	3.0	0.237 ± 0.151	0.411 ± 0.176	0.584 ± 0.230
velocity-1.6_KNN	1.0	2.0	2.0	0.389 ± 0.247	0.510 ± 0.301	0.572 ± 0.296
velocity-1.6_LR	1.0	2.0	2.0	0.352 ± 0.292	0.510 ± 0.310	0.484 ± 0.288
velocity-1.6_NB	1.0	1.0	1.0	0.441 ± 0.298	0.496 ± 0.296	0.501 ± 0.302
xalan-2.6_J48	1.0	2.0	3.0	0.374 ± 0.289	0.535 ± 0.302	0.604 ± 0.314
xalan-2.6_KNN	1.0	2.0	3.0	0.260 ± 0.232	0.533 ± 0.211	0.606 ± 0.255
xalan-2.6_LR	1.0	3.0	2.0	0.472 ± 0.297	0.669 ± 0.295	0.535 ± 0.279
xalan-2.6_NB	1.0	2.0	2.0	0.459 ± 0.312	0.572 ± 0.339	0.525 ± 0.323
xerces-1.4_J48	1.0	1.0	1.0	0.525 ± 0.318	0.581 ± 0.344	0.587 ± 0.352
xerces-1.4_KNN	1.0	3.0	2.0	0.304 ± 0.285	0.450 ± 0.286	0.384 ± 0.276
xerces-1.4_LR	1.0	3.0	2.0	0.398 ± 0.262	0.527 ± 0.263	0.456 ± 0.250
xerces-1.4_NB	1.0	1.0	1.0	0.527 ± 0.317	0.529 ± 0.318	0.527 ± 0.317
Average	**1.00**	**2.05**	**2.17**	**0.431 ± 0.261**	**0.559 ± 0.267**	**0.573 ± 0.278**

Table 5: Detailed instance-level results of MMO_b, MMO_o, and PMO for Software Product Lines Testing, showing Scott-Knott ESD ranks and the mean ± standard deviation of optimized and normalized performance (smaller is better).

Instance	**Rank**			**Mean ± Std**		
	MMO_b	MMO_o	PMO	MMO_b	MMO_o	PMO
7z	1.0	2.0	3.0	0.100 ± 0.086	0.133 ± 0.070	0.517 ± 0.228
amazon	1.0	2.0	3.0	0.228 ± 0.115	0.308 ± 0.159	0.719 ± 0.195
berkeleydbc	1.0	2.0	3.0	0.114 ± 0.090	0.229 ± 0.120	0.514 ± 0.254
cocheecologico	1.0	2.0	3.0	0.075 ± 0.069	0.229 ± 0.139	0.765 ± 0.150
counterstrikesimplefeatu	1.0	2.0	3.0	0.064 ± 0.063	0.171 ± 0.108	0.579 ± 0.195
drupal	1.0	2.0	3.0	0.166 ± 0.071	0.370 ± 0.177	0.787 ± 0.141
dssample	1.0	2.0	3.0	0.100 ± 0.082	0.167 ± 0.159	0.633 ± 0.235
dune	1.0	2.0	3.0	0.188 ± 0.135	0.425 ± 0.134	0.713 ± 0.145
electronicdrum	1.0	2.0	3.0	0.239 ± 0.093	0.280 ± 0.131	0.764 ± 0.183
hipacc	1.0	2.0	3.0	0.170 ± 0.086	0.345 ± 0.109	0.664 ± 0.265
javagc	1.0	2.0	3.0	0.100 ± 0.057	0.304 ± 0.131	0.624 ± 0.178
jhipster	1.0	2.0	3.0	0.148 ± 0.085	0.245 ± 0.157	0.561 ± 0.228
lrzip	1.0	1.0	1.0	0.500 ± 0.000	0.500 ± 0.000	0.500 ± 0.000
modeltransformation	1.0	2.0	3.0	0.182 ± 0.112	0.295 ± 0.099	0.645 ± 0.223
polly	1.0	2.0	3.0	0.215 ± 0.131	0.365 ± 0.163	0.755 ± 0.185
smarthomev2.2	1.0	2.0	3.0	0.098 ± 0.064	0.284 ± 0.143	0.761 ± 0.144
splssimuelespnp	1.0	2.0	3.0	0.156 ± 0.078	0.317 ± 0.064	0.656 ± 0.237
videoplayer	1.0	2.0	3.0	0.167 ± 0.096	0.333 ± 0.148	0.854 ± 0.108
vp9	1.0	2.0	3.0	0.276 ± 0.114	0.428 ± 0.107	0.740 ± 0.146
webportal	1.0	2.0	3.0	0.109 ± 0.072	0.222 ± 0.089	0.598 ± 0.251
x264	1.0	2.0	3.0	0.075 ± 0.121	0.275 ± 0.142	0.575 ± 0.237
Average	**1.00**	**1.95**	**2.90**	**0.165 ± 0.087**	**0.296 ± 0.121**	**0.663 ± 0.187**

Table 6: Detailed instance-level results of MMO_b, MMO_o, and PMO for Software Project Scheduling, showing Scott-Knott ESD ranks and the mean ± standard deviation of optimized and normalized performance (smaller is better).

Instance	**Rank**			**Mean ± Std**		
	MMO_b	MMO_o	PMO	MMO_b	MMO_o	PMO
10-10-skill-4-5	1.0	2.0	3.0	0.007 ± 0.005	0.013 ± 0.007	0.841 ± 0.110
10-10-skill-6-7	1.0	2.0	3.0	0.011 ± 0.006	0.013 ± 0.005	0.749 ± 0.136
10-15-skill-4-5	1.0	2.0	3.0	0.004 ± 0.001	0.005 ± 0.003	0.907 ± 0.064
10-15-skill-6-7	1.0	2.0	3.0	0.001 ± 0.001	0.004 ± 0.003	0.904 ± 0.065
10-5-skill-4-5	1.0	1.0	2.0	0.000 ± 0.000	0.001 ± 0.002	0.772 ± 0.113
10-5-skill-6-7	1.0	2.0	3.0	0.002 ± 0.002	0.023 ± 0.044	0.612 ± 0.162
20-10-skill-4-5	1.0	2.0	3.0	0.006 ± 0.004	0.011 ± 0.008	0.843 ± 0.066
20-10-skill-6-7	1.0	2.0	3.0	0.004 ± 0.003	0.005 ± 0.004	0.853 ± 0.085
20-15-skill-4-5	1.0	2.0	3.0	0.011 ± 0.006	0.014 ± 0.008	0.872 ± 0.065
20-15-skill-6-7	1.0	2.0	3.0	0.016 ± 0.016	0.032 ± 0.015	0.940 ± 0.048
20-5-skill-4-5	1.0	2.0	3.0	0.011 ± 0.007	0.022 ± 0.016	0.807 ± 0.120
20-5-skill-6-7	1.0	2.0	3.0	0.006 ± 0.004	0.016 ± 0.004	0.826 ± 0.123
30-10-skill-4-5	1.0	2.0	3.0	0.006 ± 0.004	0.011 ± 0.007	0.855 ± 0.075
30-10-skill-6-7	1.0	2.0	3.0	0.007 ± 0.004	0.010 ± 0.007	0.880 ± 0.068
30-15-skill-4-5	1.0	2.0	3.0	0.008 ± 0.004	0.020 ± 0.015	0.874 ± 0.096
30-15-skill-6-7	1.0	2.0	3.0	0.013 ± 0.009	0.027 ± 0.012	0.902 ± 0.057
30-5-skill-4-5	1.0	2.0	3.0	0.016 ± 0.008	0.036 ± 0.016	0.831 ± 0.091
30-5-skill-6-7	1.0	2.0	3.0	0.004 ± 0.004	0.021 ± 0.018	0.874 ± 0.110
Average	**1.00**	**1.94**	**2.94**	**0.007 ± 0.005**	**0.016 ± 0.011**	**0.841 ± 0.092**

Table 7: Detailed instance-level results of MMO_b, MMO_o, and PMO for Third Party Library Migration, showing Scott-Knott ESD ranks and the mean ± standard deviation of optimized and normalized performance (smaller is better).

Instance	**Rank**			**Mean ± Std**		
	MMO_b	MMO_o	PMO	MMO_b	MMO_o	PMO
migrationRule10_ALL	2.0	3.0	1.0	0.000 ± 0.000	0.396 ± 0.512	0.000 ± 0.000
migrationRule10_CO	1.0	1.0	1.0	0.000 ± 0.000	0.100 ± 0.316	0.100 ± 0.316
migrationRule10_DS	2.0	3.0	1.0	0.000 ± 0.000	0.224 ± 0.412	0.000 ± 0.000
migrationRule10_MS	1.0	1.0	1.0	0.500 ± 0.000	0.500 ± 0.000	0.500 ± 0.000
migrationRule18_ALL	1.0	2.0	3.0	0.000 ± 0.000	0.064 ± 0.136	0.404 ± 0.264
migrationRule18_CO	1.0	1.0	1.0	0.500 ± 0.000	0.500 ± 0.000	0.500 ± 0.000
migrationRule18_DS	1.0	1.0	1.0	0.500 ± 0.000	0.500 ± 0.000	0.500 ± 0.000
migrationRule18_MS	1.0	1.0	2.0	0.000 ± 0.000	0.017 ± 0.053	0.217 ± 0.284
migrationRule1_ALL	1.0	2.0	3.0	0.020 ± 0.020	0.140 ± 0.152	0.664 ± 0.228
migrationRule1_CO	1.0	2.0	3.0	0.011 ± 0.015	0.113 ± 0.146	0.623 ± 0.282
migrationRule1_DS	1.0	1.0	1.0	0.500 ± 0.000	0.500 ± 0.000	0.500 ± 0.000
migrationRule1_MS	1.0	2.0	3.0	0.000 ± 0.000	0.080 ± 0.169	0.760 ± 0.207
migrationRule2_ALL	1.0	2.0	3.0	0.049 ± 0.050	0.121 ± 0.117	0.881 ± 0.093
migrationRule2_CO	1.0	2.0	3.0	0.078 ± 0.078	0.151 ± 0.128	0.793 ± 0.135
migrationRule2_DS	1.0	2.0	3.0	0.067 ± 0.043	0.143 ± 0.091	0.850 ± 0.087
migrationRule2_MS	1.0	1.0	2.0	0.096 ± 0.065	0.104 ± 0.056	0.804 ± 0.139
migrationRule3_ALL	1.0	2.0	3.0	0.091 ± 0.139	0.441 ± 0.141	0.698 ± 0.222
migrationRule3_CO	1.0	2.0	2.0	0.271 ± 0.238	0.819 ± 0.160	0.796 ± 0.150
migrationRule3_MS	1.0	2.0	3.0	0.025 ± 0.079	0.225 ± 0.184	0.475 ± 0.275
migrationRule4_ALL	1.0	2.0	3.0	0.010 ± 0.031	0.120 ± 0.096	0.704 ± 0.173
migrationRule4_CO	1.0	1.0	2.0	0.000 ± 0.000	0.000 ± 0.000	1.000 ± 0.000
migrationRule4_DS	1.0	1.0	1.0	0.500 ± 0.000	0.500 ± 0.000	0.500 ± 0.000
migrationRule4_MS	1.0	1.0	2.0	0.000 ± 0.000	0.067 ± 0.211	0.600 ± 0.377
migrationRule5_ALL	1.0	3.0	2.0	0.070 ± 0.154	0.348 ± 0.292	0.203 ± 0.133
migrationRule5_CO	1.0	2.0	2.0	0.000 ± 0.000	0.350 ± 0.337	0.400 ± 0.211
migrationRule5_DS	1.0	3.0	2.0	0.000 ± 0.000	0.259 ± 0.280	0.195 ± 0.345
migrationRule5_MS	1.0	1.0	1.0	0.000 ± 0.000	0.100 ± 0.316	0.100 ± 0.316
migrationRule7_ALL	1.0	2.0	3.0	0.136 ± 0.076	0.282 ± 0.182	0.849 ± 0.144
migrationRule7_CO	1.0	2.0	3.0	0.106 ± 0.078	0.240 ± 0.134	0.852 ± 0.127
migrationRule7_DS	1.0	2.0	3.0	0.159 ± 0.069	0.258 ± 0.069	0.826 ± 0.151
migrationRule7_MS	1.0	2.0	3.0	0.091 ± 0.061	0.164 ± 0.118	0.764 ± 0.148
migrationRule8_ALL	1.0	1.0	2.0	0.000 ± 0.000	0.000 ± 0.000	0.617 ± 0.209
migrationRule8_CO	1.0	2.0	2.0	0.000 ± 0.000	0.000 ± 0.000	0.200 ± 0.422
migrationRule8_DS	1.0	1.0	1.0	0.500 ± 0.000	0.500 ± 0.000	0.500 ± 0.000
migrationRule8_MS	1.0	2.0	2.0	0.000 ± 0.000	0.000 ± 0.000	0.400 ± 0.516
Average	**1.06**	**1.74**	**2.11**	**0.122 ± 0.034**	**0.238 ± 0.137**	**0.536 ± 0.170**

Table 8: Detailed instance-level results of MMO_b, MMO_o, and PMO for Workflow Scheduling, showing Scott-Knott ESD ranks and the mean ± standard deviation of optimized and normalized performance (smaller is better).

Instance	**Rank**			**Mean ± Std**		
	MMO_b	MMO_o	PMO	MMO_b	MMO_o	PMO
CyberShake_100	1.0	2.0	3.0	0.011 ± 0.007	0.064 ± 0.028	0.447 ± 0.266
CyberShake_1000	1.0	2.0	3.0	0.125 ± 0.095	0.291 ± 0.120	0.779 ± 0.201
CyberShake_30	1.0	2.0	3.0	0.014 ± 0.017	0.046 ± 0.051	0.483 ± 0.276
CyberShake_50	1.0	2.0	3.0	0.020 ± 0.018	0.096 ± 0.033	0.436 ± 0.299
Epigenomics_100	1.0	2.0	3.0	0.094 ± 0.078	0.221 ± 0.170	0.294 ± 0.265
Epigenomics_24	1.0	2.0	3.0	0.012 ± 0.007	0.027 ± 0.006	0.157 ± 0.307
Epigenomics_46	1.0	2.0	3.0	0.036 ± 0.051	0.249 ± 0.199	0.631 ± 0.246
Epigenomics_997	1.0	2.0	3.0	0.196 ± 0.131	0.365 ± 0.205	0.760 ± 0.177
Inspiral_100	1.0	2.0	3.0	0.041 ± 0.028	0.110 ± 0.053	0.658 ± 0.235
Inspiral_1000	1.0	2.0	3.0	0.131 ± 0.097	0.344 ± 0.153	0.736 ± 0.186
Inspiral_30	1.0	2.0	3.0	0.000 ± 0.000	0.000 ± 0.000	0.512 ± 0.248
Inspiral_50	1.0	2.0	3.0	0.030 ± 0.030	0.090 ± 0.046	0.691 ± 0.213
Montage_100	1.0	2.0	3.0	0.039 ± 0.032	0.087 ± 0.039	0.764 ± 0.201
Montage_1000	1.0	2.0	3.0	0.169 ± 0.149	0.380 ± 0.282	0.836 ± 0.107
Montage_25	1.0	2.0	3.0	0.015 ± 0.007	0.079 ± 0.023	0.563 ± 0.273
Montage_50	1.0	2.0	3.0	0.018 ± 0.019	0.061 ± 0.031	0.395 ± 0.246
Sipht_100	1.0	2.0	3.0	0.000 ± 0.000	0.003 ± 0.003	0.294 ± 0.326
Sipht_1000	1.0	2.0	3.0	0.061 ± 0.076	0.259 ± 0.129	0.694 ± 0.190
Sipht_30	1.0	2.0	3.0	0.005 ± 0.005	0.018 ± 0.012	0.150 ± 0.299
Sipht_60	1.0	2.0	3.0	0.003 ± 0.002	0.064 ± 0.073	0.175 ± 0.298
Average	**1.00**	**2.00**	**3.00**	**0.051 ± 0.042**	**0.143 ± 0.083**	**0.523 ± 0.243**

Table 9: Detailed instance-level results of MMO_b, MMO_o, and PMO for Web Service Composition, showing Scott-Knott ESD ranks and the mean ± standard deviation of optimized and normalized performance (smaller is better).

Instance	**Rank**			**Mean ± Std**		
	MMO_b	MMO_o	PMO	MMO_b	MMO_o	PMO
workflow_1	1.0	2.0	3.0	0.096 ± 0.069	0.165 ± 0.111	0.665 ± 0.165
workflow_10	1.0	2.0	3.0	0.133 ± 0.078	0.293 ± 0.159	0.533 ± 0.180
workflow_10_reverse	1.0	2.0	3.0	0.361 ± 0.181	0.522 ± 0.096	0.686 ± 0.268
workflow_1_reverse	1.0	2.0	3.0	0.477 ± 0.198	0.690 ± 0.176	0.743 ± 0.124
workflow_2	1.0	2.0	3.0	0.128 ± 0.086	0.200 ± 0.128	0.657 ± 0.247
workflow_2_reverse	1.0	2.0	3.0	0.444 ± 0.197	0.689 ± 0.133	0.807 ± 0.112
workflow_3	1.0	2.0	3.0	0.193 ± 0.140	0.401 ± 0.113	0.704 ± 0.206
workflow_3_reverse	1.0	2.0	3.0	0.403 ± 0.174	0.708 ± 0.138	0.752 ± 0.179
workflow_4	1.0	2.0	3.0	0.081 ± 0.065	0.250 ± 0.195	0.616 ± 0.286
workflow_4_reverse	1.0	2.0	3.0	0.317 ± 0.181	0.441 ± 0.192	0.701 ± 0.187
workflow_5	1.0	2.0	3.0	0.150 ± 0.108	0.335 ± 0.175	0.743 ± 0.163
workflow_5_reverse	1.0	2.0	2.0	0.309 ± 0.194	0.640 ± 0.230	0.678 ± 0.142
workflow_6	1.0	2.0	3.0	0.095 ± 0.075	0.175 ± 0.166	0.629 ± 0.232
workflow_6_reverse	1.0	2.0	3.0	0.393 ± 0.187	0.572 ± 0.194	0.744 ± 0.169
workflow_7	1.0	2.0	3.0	0.170 ± 0.122	0.347 ± 0.181	0.725 ± 0.205
workflow_7_reverse	1.0	2.0	3.0	0.357 ± 0.210	0.667 ± 0.190	0.804 ± 0.140
workflow_8	1.0	2.0	3.0	0.164 ± 0.096	0.321 ± 0.129	0.601 ± 0.204
workflow_8_reverse	1.0	2.0	3.0	0.297 ± 0.205	0.590 ± 0.109	0.738 ± 0.153
workflow_9	1.0	2.0	3.0	0.152 ± 0.117	0.357 ± 0.148	0.704 ± 0.241
workflow_9_reverse	1.0	2.0	2.0	0.353 ± 0.203	0.692 ± 0.180	0.650 ± 0.218
Average	**1.00**	**2.00**	**2.90**	**0.254 ± 0.144**	**0.453 ± 0.157**	**0.694 ± 0.191**